Technology Innovator in residence (TIR) Specific Questions

This comprehensive guide outlines all questions within the application that require a selection from predefined options or structured long-form input, including the exact choices and thematic groupings available to the applicant.

1. Professional Profile

1.1 LinkedIn or Professional Profile URL [Lead & Co-founders]

Input Type:	URL
Required:	Optional
Evaluates:	Professional network analysis for background verification, skill endorsement validation, and professional trajectory assessment. Profiles with <100 connections or <3 skill endorsements in relevant domains trigger additional verification steps. Other red and green flags can be identified and coded.
Example Answer:	e.g., linkedin.com/in/arunkumar
Conversational Prompt:	
Status Check	

1.2 Upload your CV/Resume [Lead & Co-founders]

Input Type:	File Upload (PDF, max 5MB)
Required:	Required
Evaluates:	Comprehensive profile extraction. Our CV parser auto-populates fields (education, experience, skills, publications, patents, projects) reducing form completion time by 60%. All fields are highlighted for manual review.
Example Answer:	PDF format only. Upload to auto-fill sections below.
Conversational Prompt:	
Status Check	

1.3 GitHub Profile URL [Lead & Co-founders]

Input Type:	File Upload (PDF, max 5MB)
Required:	Optional
Evaluates:	Comprehensive profile extraction. Our CV parser auto-populates 12 fields (education, experience, skills, publications, patents, projects) reducing form completion time by 60%. Parsed data is flagged with confidence scores; low-confidence fields are highlighted for manual review.

Example Answer:	PDF format only. Upload to auto-fill sections below.

2. Basic Information

✦ Upload your CV to auto-fill sections marked with purple indicator

2.1 Do you have a team?

Input Type:	Single-select (Yes / No)
Required:	Required
Evaluates:	Application routing (solo vs. team flow). If Yes, repeats certain sections for onboarding co-founders
Example Answer:	e.g., Dr. Arun Kumar
Conversational Prompt:	
Status Check	

2.2 Full Name [CV Auto-Fill] [Lead & Co-founders]

Input Type:	Short Text
Required:	Required
Evaluates:	Identity verification
Example Answer:	e.g., Dr. Arun Kumar
Conversational Prompt:	
Status Check	

2.3 Phone Number [CV Auto-Fill] [Lead & Co-founders]

Input Type:	Phone
Required:	Required
Evaluates:	Contact Verification and Future reference
Example Answer:	e.g., +91 98765 43210
Conversational Prompt:	
Status Check	

2.4 Email [CV Auto-Fill] [Lead & Co-founders]

Input Type:	Email
Required:	Required
Evaluates:	Primary Contact Channel, Login profile anchor

Example Answer:	e.g., arun.kumar@example.com
Conversational Prompt:	
Status Check	

2.5 Current Organization [CV Auto-Fill] [Lead & Co-founders]

Input Type:	Short Text
Required:	Required
Evaluates:	Institutional quality signal for research infrastructure assessment. Tier-1 institutions (IITs, IISc, NITs, top global universities) indicate exposure to rigorous research environments. However, we actively seek outliers from non-tier-1 institutions who demonstrate exceptional self-driven research capability.
Example Answer:	e.g., IISc Bangalore / Independent / Company Name
Conversational Prompt:	
Status Check	

2.6 Highest Technology Degree Achieved [CV Auto-Fill] [Lead & Co-founders]

O	**Select one:**
○	Bachelor's Degree
○	Master's Degree
○	PhD
Input Type:	Single Select
Required:	Required
Evaluates:	Academic credential baseline. PhD holders in relevant domains show 34% higher venture success rate in ARTPARK cohorts. However, exceptional self-taught engineers with demonstrated builds (GitHub, shipped products) receive equivalent evaluation weight. We use this to calibrate expected research output during residency.
Example Answer:	Select one:
Conversational Prompt:	
Status Check	

2.7 What are other incubators / accelerators have you been a part of (in the past and currently)?

Input Type:	Short Text (or "None")
Required:	Required
Evaluates:	Conflict of commitment, existing support structure. Also DST does not allow incubating companies to be funded from multiple sources of DST Grants
Example Answer:	e.g., None / IIT Madras Incubation Cell
Conversational Prompt:	
Status Check	

2.8 How did you hear about ARTPARK?

	Select one:
○	Referral from friend/colleague
○	IISc faculty or staff
○	Social media (LinkedIn, Twitter, etc.)
○	Event or conference
○	Search engine
○	Partner organization
○	News article or press
○	Other
Input Type:	Single Select
Required:	Optional Required
Evaluates:	Helps us track our outreach effectiveness
Example Answer:	Select one:
Conversationa l Prompt:	
Status Check	

3. Problem and its importance

3.1 Is the problem you want to solve well-defined? ?

O	Select one:
○	Yes, clearly defined (describe below)
○	Partially defined
○	Still exploring the problem space
Input Type:	Single Select
Required:	Required
Evaluates:	Evaluate the maturity and base condition of the innovator starting in a domain of operation. Generally innovators who are passionate would have strong sense of certain problem areas
Example Answer:	Yes, clearly defined

3.2 Describe the problem you want to solve? Why is now the right time to solve it?

Be as specific as possible.

Input Type:	Long Text (1000 chars)
Required:	Conditional Required, if the above response says partially or clearly defined problem
Evaluates:	Provides high level understanding of the problem areas which the founder wishes to operate within
Example Answer:	Smallholder farmers lack access to real-time soil intelligence, leading to 20–30% yield loss despite increased fertilizer usage.

3.3 Why is this an important problem to solve?

Quantification of the problem size or market is useful.

Input Type:	Long Text (1000 chars)
Required:	Conditional Required, if the above response says partially or clearly defined problem
Evaluates:	Helps understand the market size, urgency and impact
Example Answer:	India has 140M+ hectares of farmland. Inefficient input usage leads to ₹1.2L Cr annual losses and environmental degradation.

4. Your solution to the problem

4.1 How far along are you in developing your solution?

O	Select one:
○	Still exploring problem area

○	Select one:
○	Literature / research stage
○	Simulations completed
○	Lab demos done as proof-of-concept
○	Prototype built
○	Pilot-ready product
○	Deployed in real setting with real users
Input Type:	Single Select
Required:	Required
Evaluates:	Development stage assessment for residency planning. Early-stage applicants (exploring/literature) need a 6-month problem definition phase. Mid-stage (lab validated/prototype) can enter directly into the commercialization track. Late-stage (deployed) may qualify for fast-track venture spin-out rather than standard IIR
Example Answer:	Select one:

4.2 Explain your proposed solution clearly and how it connects to the problem.

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Solution clarity, logical connection to problem
Example Answer:	We are building a low-cost IoT soil sensor network with AI-based nutrient prediction, enabling farmers to optimize fertilizer usage in real time.

4.3 What is the core technology innovation at the heart of your solution? Describe the specific laboratory-proven research or cutting-edge advance you intend to translate.

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Technical depth, innovation
Example Answer:	Magnetic induction sensor and algorithms that are more accurate because of X technological differentiation.

4.4 How does this represent a 10x improvement over existing state-of-the-art solutions rather than an incremental gain?

Input Type:	Long Text (1500 chars)

Required:	Required
Evaluates:	Breakthrough vs incremental innovation
Example Answer:	Reduces sensor cost from ₹15,000 to ₹1,200 while improving accuracy by 3x through adaptive calibration.

4.5 What are the primary technical hurdles you must solve to make your technology robust for real-world deployment?

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Real-world engineering awareness
Example Answer:	Handling soil variability, temperature drift, and sensor degradation over long-term deployment.

4.6 What is the “unfair advantage” of your technology? Why is it difficult to replicate?

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Defensibility and long-term advantage
Example Answer:	Proprietary dataset from 10,000+ soil samples combined with patented sensor architecture.

4.7 How does your innovation contribute to national transformation, and what is your roadmap to becoming globally competitive?

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Scale thinking and ambition
Example Answer:	Starting with India’s agri sector, expanding to Southeast Asia and Africa with similar soil conditions.

4.8 Describe your potential Customers? Why will they pay for this solution?

Describe the economic or operational value proposition.

Input Type:	Long Text (1500 chars)
Required:	Required
Evaluates:	Business viability and customer value

Example Answer:	Farmers save ₹5,000 per acre annually on fertilizer, achieving ROI within one crop cycle.

5. Execution Plan

5.1 What are the top 2–3 things that will break or fail when moving from the lab to real-world deployment?

Input Type:	Long Text (max 1000 characters)
Required:	Required
Evaluates:	Practical risk awareness and mitigation thinking
Example Answer:	Sensor calibration drift, connectivity issues in rural areas, and physical wear-and-tear.

5.2 What is the most critical milestone during this residency? What does success look like?

Input Type:	Long Text (max 1000 characters)
Required:	Required
Evaluates:	Milestone clarity and execution focus
Example Answer:	Deploy 100 pilot units across 3 states with validated performance benchmarks.

5.3 Budget with justification and milestones for the current year

Input Type:	Long Text (max 2000 characters) or File Upload (PDF/XLS)
Required:	Required
Evaluates:	Financial planning realism and milestone discipline. Reviewers look for quarterly milestones tied to line-item spend, not vague annual totals. Over-allocation to salaries with no customer-facing milestones is a red flag; so is an unrealistically small budget for the proposed work.
Example Answer:	e.g., Total ₹45L over 12 months. Q1–Q2 (₹20L): hardware R&D and component procurement for 100 units. Q3 (₹15L): pilot deployment across 50 MSME units in Coimbatore and Ludhiana. Q4 (₹10L): team salaries (2 engineers) and BIS certification prep. Milestones: 10 paid pilots by Month 6, validated unit economics by Month 9, seed round closed by Month 12.

5.4 Translating deep tech is notoriously difficult and prone to failure. Describe a time a research direction or prototype failed significantly. How did you pivot, and what did it teach you about the path to commercialization?

Input Type:	Long Text (500 chars)
Required:	Required
Rationale:	Risk tolerance and delayed gratification assessment. Research and deep-tech innovation require sustained effort without immediate rewards. Candidates with prior experience in open-ended projects (PhD research, open-source maintenance, long-term volunteer work) demonstrate psychological readiness for residency demands. We verify claims through reference checks.
Example Answer:	

6. Evidence

6.1 Please share evidence related to your proposed solution (publications, patent applied/granted, photos of your prototype/product, etc.)

Input Type:	File upload (multi-file)
Required:	Optional
Evaluates:	Evidence of some tangible work done by the innovator
Example Answer:	prototype_v2.jpg, patent_application.pdf

6.2 Please share a video (shorter than 3 minutes) of your prototype or product to help understand the technology and its status of development.

Input Type:	File upload (multi-file)
Required:	Optional
Evaluates:	Working-state verification. Photos can be staged; video of the product running, measurements being taken, or a demo interaction shows actual functionality. Also surfaces the applicant's ability to communicate technical work visually
Example Answer:	https://www.loom.com/share/abc123 — 2:45 minute walkthrough showing sensor installation and dashboard view from a live deployment

6.3 Please upload a 6-slide presentation that would cover the basic pitch of the proposed solution.

Input Type:	Upload (PDF/PPT, max 10MB)
Required:	Required
Evaluates:	Evaluates: Structured communication under constraint. A 6-slide format forces prioritization: problem, solution, core technology, customers, milestones, ask. Tests pitch hygiene and the ability to compress a complex story into a decision-ready format. Expected sections: (1) Problem, (2) Solution, (3) Core Technology, (4) Market & Customers, (5) Milestones & Budget, (6) Why You.
Example Answer:	ProjectPitch_ARTPARK_v3.pdf

7. Declaration

7.1 I confirm my information submitted by me within the documentation above is true and completely relevant in terms of the questions required

Input Type:	Checkbox
Required:	Required
Evaluates:	Submission integrity
Example Answer:	Checked

7.2 I consent to reference checks

Input Type:	Checkbox
Required:	Required
Evaluates:	Verification permission
Example Answer:	Checked

7.3 I agree to program terms and data policy

Input Type:	Checkbox
Required:	Required
Evaluates:	Terms acceptance
Example Answer:	Checked

7.4 I agree to receive newsletters and future communication from ARTPARK

Input Type:	Checkbox
Required:	Required
Evaluates:	Terms acceptance
Example Answer:	Checked